CPU Scheduling: Textbook Excerpt

Operating Systems | Chapter 5 | Category: Textbook

5.1 Scheduling in multiprogramming

CPU scheduling is central to multiprogramming because several processes may be ready to run while only one can use a single CPU core at a time. The operating system chooses a process from the ready queue to improve utilization and service quality.

Scheduling decisions can occur when a process changes from running to waiting, running to ready, waiting to ready, or when a process terminates. Algorithms that can interrupt a running process are called preemptive; algorithms that let it continue until it blocks or finishes are non-preemptive.

5.2 Measuring performance

CPU utilization measures how often the CPU is doing useful work. Throughput measures completed processes per time unit. Turnaround time is completion time minus arrival time. Waiting time adds the time spent waiting in the ready queue, and response time measures delay until the first response.

Minimizing average waiting time is useful for batch workloads. Interactive workloads also value predictable response time and fairness. A single scheduling algorithm cannot optimize every criterion simultaneously.

5.3 FCFS and SJF

First-Come, First-Served schedules processes in order of arrival and is easy to implement with a FIFO queue. Its principal disadvantage is the convoy effect: short processes may wait behind one long CPU-bound process.

Shortest Job First chooses the process whose next CPU burst is shortest. If burst lengths are known, SJF is optimal for minimizing average waiting time among non-preemptive choices. In practice burst length is estimated, and long processes can starve if short work continues to arrive.

Shortest Remaining Time First is the preemptive counterpart. When a new process becomes ready with a shorter remaining CPU burst than the running process, the scheduler may switch to the new process.

5.4 Round Robin

Round Robin assigns each ready process a time quantum, often a few milliseconds. When the quantum expires, the scheduler places an unfinished process at the end of the ready queue and runs the next process. This gives users regular opportunities to receive CPU service.

If the quantum is very large, Round Robin behaves much like FCFS. If the quantum is too small, context-switching overhead becomes excessive. Choosing a practical quantum is therefore a trade-off between response time and overhead.

5.5 Priority and starvation

Priority scheduling chooses the ready process with the highest configured priority. A disadvantage is starvation, where a low-priority process waits indefinitely. Aging is a remedy: a process gains priority gradually the longer it waits.

For exam questions, state the assumptions explicitly: arrival times, CPU bursts, priorities, preemption policy, and Round Robin quantum. Draw the Gantt chart before calculating average waiting or turnaround time.